MIT 8103 – ADVANCED DATABASE SYSTEMS

Final Portfolio Documentation

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Case Study	E-commerce Management Platform

1. Introduction

This project explores the design, implementation, optimisation and management of databases using an e-commerce management platform as a case study. The platform manages customers, product categories, products, inventory, customer orders, order items, payments and deliveries. PostgreSQL was used for relational database activities, while MongoDB was used to explore document-oriented modelling and distributed database concepts.

2. Summary of Portfolio Activities

Portfolio 1 – Database Design and Modelling

A relational database was designed and implemented using PostgreSQL 16.15. It contains eight main tables: Customer, Category, Product, Inventory, Orders, Order_Item, Payment and Delivery. Primary keys, foreign keys, unique constraints and check constraints were applied to support data integrity. An Entity Relationship Diagram (ERD) was created, sample records were inserted, and JOIN queries were used to retrieve customer, order, payment, delivery and product information.

Portfolio 2 – Query Processing and Optimisation

SQL queries were analysed with `EXPLAIN ANALYZE`. Indexes were created on the orders customer identifier and on the inventory low-stock condition. Execution plans were compared before and after indexing. The results showed that an index does not always lead to an index scan: for small tables, PostgreSQL may choose a sequential scan because it is estimated to be more efficient.

Portfolio 3 – Transactions and Concurrency

Transaction control was explored with `BEGIN`, `COMMIT` and `ROLLBACK`. A rollback test confirmed that an uncommitted inventory update was reversed, while a commit test demonstrated that an update could be saved. The default transaction isolation level was inspected and found to be Read Committed. **Outstanding work:** the two-session concurrency/isolation demonstration still needs to be completed before this portfolio can be described as fully complete.

Portfolio 4 – NoSQL and Advanced Data Models

MongoDB Community Edition 8.0.34 was installed on Ubuntu and the `ecommerce_nosql` database was created. A document-oriented order model was explored, with related customer, product, payment and delivery information represented within order documents. This can make related information convenient to retrieve, although embedding may introduce duplication and update-management trade-offs. The MongoDB scripts and query evidence should be checked and added to the repository before final submission.

Portfolio 5 – Distributed and Cloud Databases

MongoDB was configured as a replica set named `rs0`, and its primary member status was inspected. A proposed shard key of `customer_id` was used in a logical partitioning simulation. Update/read operations and an insert using majority write concern were verified. **Implementation limitation:** the replica set contains one member, so the exercise does not demonstrate cross-server replication or automatic failover. The sharding exercise is a simulation, not a deployed sharded cluster.

3. Tools and Technologies

Technology	Use in the project
PostgreSQL 16.15	Relational schema, SQL queries, constraints, indexes and transactions.
MongoDB 8.0.34	Document modelling, replica set configuration and sharding concepts.
mongosh	MongoDB database operations and verification.
Ubuntu Linux	Local development and database environment.
Git and GitHub	Version control, project organisation and documentation.

4. Challenges and Lessons Learned

A key challenge was understanding how indexes influence query execution. The practical results showed that indexing does not guarantee an index scan, because the query planner considers the amount of data and estimated cost. Transaction exercises clarified the difference between committing and rolling back changes. Working with MongoDB also highlighted the differences between relational tables and document-oriented structures. Configuring a local replica set provided exposure to distributed database concepts while also showing the limits of a single-node environment.

5. Conclusion

The project provided practical experience in database design, SQL querying, query-plan analysis, transaction control, NoSQL modelling and introductory distributed database configuration. It demonstrated how relational and document databases can address different data-management needs. Further improvements include completing the two-session concurrency experiment, verifying and documenting the Portfolio 4 queries, deploying a multi-node replica set and testing a real sharded or managed cloud database environment.

6. Repository, References and AI Use Declaration

GitHub repository: https://github.com/MEENAH0557/mit-8103-advanced-database-portfolio

References

1. PostgreSQL Documentation: https://www.postgresql.org/docs/
2. MongoDB Manual: https://www.mongodb.com/docs/manual/
3. MongoDB Replication Documentation: https://www.mongodb.com/docs/manual/replication/
4. MongoDB Sharding Documentation: https://www.mongodb.com/docs/manual/sharding/

AI Use Declaration

ChatGPT was used as a learning and support tool for explanations of database concepts, troubleshooting, SQL and MongoDB guidance, and assistance organising documentation. Practical database operations and verification were performed by the student in the local development environment. AI assistance was not treated as a substitute for understanding or verifying the results.